

AMAN KARN

Senior Game & Software Engineer

Unity3D | Unreal Engine 5 | C# | C++ | VR / AR / MR | Game Architecture

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PROFESSIONAL SUMMARY

Senior Game & Software Engineer with over 5 years of experience building real-time interactive games and immersive applications using Unity3D and Unreal Engine 5. Proven expertise across 2D, 3D, VR, AR, and MR platforms, delivering 30+ applications and complex gameplay systems. Experienced in architecting modular codebase structures, converting classic gameplay into spatial XR experiences, generating synthetic datasets for machine learning, and mentoring engineering teams. Strong technical background with a Bachelor's in Computer Engineering.

TECHNICAL SKILLS

Game Engines:	Unity3D (5+ yrs), Unreal Engine 5 (1 yr)
Languages:	C, C++, C#, Python, Rust, Java, Js, Golang, PowerShell
XR Platforms:	Oculus Go / Rift / Quest 1 & 2, Magic Leap One, HoloLens One, AR/VR/MR
Architecture & Systems:	Modular & Scalable Design, Gameplay Systems, UI Systems, Synthetic Data
Tools & Workflows:	Git, GitHub, AWS, AWS CodeCommit, CDN, Jira, GitLab, Visual Studio, Blender

PROFESSIONAL EXPERIENCE

Senior Game Developer | Earth9 Pvt. Ltd. (Whitehatengineering Inc.) *Dec 2022 – Nov 2025 | Kathmandu, Nepal*

- Architected and developed scalable gameplay features and core mechanics using Unity3D and Unreal Engine 5.
- Engineered a real-world sandbox application incorporating satellite-fetched building data and real-world terrain mesh generation.
- Integrated front-end gameplay systems with back-end player achievement and progress-tracking web applications.
- Supported engineering recruitment and led the onboarding, mentoring, and training of new team members in Unreal Engine 5 gameplay workflows, Blueprint scripting, and Blender prop creation.
- Collaborated closely with UI and QA teams using Jira, Git, AWS CodeCommit, and CDN deployment workflows.

Unity3D Game Developer (Junior to Mid) | Paracosma Nepal Pvt. Ltd. *Apr 2019 – Dec 2022 | Kathmandu, Nepal*

- Developed, maintained, and tested over 30 2D, 3D, AR, and VR applications using Unity3D and C#.
- Successfully adapted and converted 9 popular 2D board games into spatial 3D/VR experiences, optimizing UI and interactions for Oculus Go, Rift, Quest 1/2, and Magic Leap One.
- Co-developed a specialized VR psycho-education and therapy application in collaboration with psychiatrists, medical personnel, and artists to support social interaction challenges.
- Built synthetic image dataset generation environments within Unity3D for machine learning safety clients; developed Python verification scripts to automate dataset quality checks.
- Developed real-time 3D camera filters and interactive stickers for video conferencing platforms (Zoom, Google Meet).

Software Developer Intern | LeapFrog Technologies Pvt. Ltd. *Sep 2018 – Oct 2018 | Kathmandu, Nepal*

- Built web interface prototypes using HTML, CSS, JavaScript, and Photoshop.
- Utilized Git version control, Jira, and Trello within an agile team setting.

FEATURED PROJECTS

Real-World Terrain Sandbox Game

Unity3D · C# · Satellite Data · Unreal Engine 5 · Web App

- Designed a sandbox experience enabling world creation on real-world terrain using satellite-fetched building data.
- Built player achievement tracking workflows connecting in-game actions to a web-based portal.

Virtual Reality Board Game Suite

Unity3D · C# · VR · Multi-Device Cross-Platform

- Converted classic titles (Snakes & Ladders, Tic-Tac-Toe, Jenga, Bingo, Call Break, Gin Rummy, Ludo, Mancala, Backgammon) into fully interactive 3D and VR titles with custom UI/UX for spatial input.

VR Psycho-Education & Therapy Platform

[Unity3D](#) · [VR](#) · [Healthcare / Medical XR](#)

- Collaborated with multidisciplinary medical teams to construct immersive therapy courses and social scenarios aimed at treating social interaction challenges.

Synthetic Dataset Generation for ML

[Unity3D](#) · [Python](#) · [Computer Vision / Safety](#)

- Developed procedural environment simulations in Unity to output computer vision training datasets, paired with Python automated verification scripts for automated quality checking.

EDUCATION & PUBLICATION

Bachelor's Degree in Computer Engineering

Advanced College of Engineering and Management (Tribhuvan University)

Graduated Jan 2019

Score: 70.34%

PUBLICATIONS

Statistic-Based Sentiment Analysis of Social Media Data — Published by: [IRJIET](#) 2018